Manufacturing features

Higher · Materials & manufacture · Worksheet

D&M WITH MR KELLY

Name _____ Class _____ Date _____

By the end of this worksheet I can…

- ☐ identify flashing, ejector marks, draft angles and injection points
- ☐ explain how ribs and webs strengthen a moulded part
- ☐ identify split lines and shear marks
- ☐ explain the purpose of bosses, location pins, ribs and webs

I DO — Worked example

Read how a full-mark answer is built

1 Explain the purpose of location pins, bosses and ribs/webs. (4 marks)

Full-mark answer each ✓ is one mark

Location pins line the two halves up accurately during assembly ✓. Bosses provide a fixing point for the screws ✓ and strengthen the area around the fixing ✓. Ribs add rigidity to thin walls without making them thick, and help the plastic flow through the mould ✓.

Why marks are not awarded

"They help it fit together and make it strong."

1 mark at most: not linked to each feature.

WE DO — Guided practice

Use the starters and word bank to help you

2 Describe how manufacturing features ensure accurate and efficient assembly of a product such as a hairdryer. (2 marks)

Sentence starters

First, …

Next, … using a …

Word bank

screws · bosses · halves · location · exactly · together · internal · Ribs/webs

Tip: 2 marks = 2 separate, developed points.

YOU DO **On your own** No notes. Check your answers afterwards.

3 Quick check: tick one box for each.

a) Why are ribs added to the inside of a plastic casing?
☐ To make it heavier ☐ Add rigidity without thick walls and hold internal parts ☐ For grip ☐ To increase cost

b) Location pins mainly help with...
☐ Accurate assembly ☐ Corrosion resistance ☐ Aesthetics ☐ Recycling

c) What is the main purpose of a boss on an injection-moulded casing?
☐ Decoration ☐ To take a screw and strengthen the fixing point ☐ To let air escape ☐ To show the recycling code

d) Thin excess plastic along the edge of a moulded part is called...
☐ Draft ☐ Flashing ☐ Ejector pin mark ☐ Rib

4 Key terms: write the letter of the matching description next to each term.

1. **Boss**	____	**A** Add strength and rigidity, help material flow, locate internal parts.
2. **Location pin**	____	**B** Positions mating parts accurately during assembly.
3. **Ribs/webs**	____	**C** Raised cylinder to receive a screw; adds strength at the fixing.
4. **Flashing**	____	**D** Thin excess material where the two halves of a mould meet.

A TV remote control casing is injection moulded in two parts.

5 Describe the purpose of a boss and of a rib on the inside of the casing. (2 marks)

6 Explain what is meant by ‘Split line’. (2 marks)

7 Explain what is meant by ‘Shear marks’. (2 marks)

Check and reflect
Mark your answers with the answer sheet. Go back to the checklist on the site and re-rate each “I can” statement above. Which one still needs work?